THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Chapter 13: **Statistics** | Class 10 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** 44.2	$\sum f_i = 50$ and $\sum f_i x_i = 2210$; Mean $= \frac{\sum f_i x_i}{\sum f_i} = \frac{2210}{50} = 44.2$
2	**(d)** 78.75	Median class 70–80: $l = 70$, $cf = 9$, $f = 4$, $h = 10$; Median $= l + \frac{\frac{N}{2} - cf}{f} \times h = 70 + \frac{12.5 - 9}{4} \times 10 = 78.75$
3	**(d)** 72.5	$l = 70$, $f_1 = 13$, $f_0 = 10$, $f_2 = 4$, $h = 10$; Mode $= l + \frac{f_1 - f_0}{2f_1 - f_0 - f_2} \times h = 70 + \frac{3}{12} \times 10 = 72.5$
4	**(b)** 30–40	It belongs to the class 30–40; Modal class = 30–40
5	**(c)** 16	$\frac{1650 + 35p}{34 + p} = 44.2$ gives $(35 - 44.2)p = -147.2$; $p = 16$
6	**(c)** 17.5	$\frac{10 + 25}{2}$
7	**(d)** 23.8	$\sum f_i = 50$ and $\sum f_i x_i = 1190$; Mean $= \frac{\sum f_i x_i}{\sum f_i} = \frac{1190}{50} = 23.8$
8	**(b)** 65	Median class 60–70: $l = 60$, $cf = 19$, $f = 12$, $h = 10$; Median $= l + \frac{\frac{N}{2} - cf}{f} \times h = 60 + \frac{25 - 19}{12} \times 10 = 65$
9	**(d)** 85.5	$l = 80$, $f_1 = 16$, $f_0 = 5$, $f_2 = 7$, $h = 10$; Mode $= l + \frac{f_1 - f_0}{2f_1 - f_0 - f_2} \times h = 80 + \frac{11}{20} \times 10 = 85.5$
10	**(b)** 30–40	It belongs to the class 30–40; Modal class = 30–40

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Class marks x_i: 5, 15, 25, 35, 45, 55; $\sum f_i = 100$ and $\sum f_i x_i = 2620$ (1); Mean $= \frac{\sum f_i x_i}{\sum f_i} = \frac{2620}{100} = 26.2$ (1).	1+1
12	Highest frequency is 16, so the modal class is 80–90; $l = 80$, $f_1 = 16$, $f_0 = 13$, $f_2 = 7$, $h = 10$ (1); Mode $= l + \frac{f_1 - f_0}{2f_1 - f_0 - f_2} \times h = 80 + \frac{3}{12} \times 10 = 82.5$ (1).	1+1
13	Cumulative frequencies: 16, 38, 58, 63, 82, 100; $N = 100$, so $\frac{N}{2} = 50$; Median class 20–30: $l = 20$, $cf = 38$, $f = 20$, $h = 10$ (1); Median $= l + \frac{\frac{N}{2} - cf}{f} \times h = 20 + \frac{50 - 38}{20} \times 10 = 26$ (1).	1+1
14	Class marks x_i: 65, 75, 85, 95, 105; $\sum f_i = 40$ and $\sum f_i x_i = 3040$ (1); Mean $= \frac{\sum f_i x_i}{\sum f_i} = \frac{3040}{40} = 76$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Take assumed mean $a = 35$; deviations $d_i = x_i - a$: -30, -20, -10, 0, 10, 20 (1); $\sum f_i = 100$ and $\sum f_i d_i = -880$ (1); Mean $= a + \frac{\sum f_i d_i}{\sum f_i} = 35 + \frac{-880}{100} = 26.2$ (1).	1+1+1
16	Cumulative frequencies: 13, 28, 37, 45, 50; $N = 50$, so $\frac{N}{2} = 25$ (1); Median class 10–20: $l = 10$, $cf = 13$, $f = 15$, $h = 10$ (1); Median $= l + \frac{\frac{N}{2} - cf}{f} \times h = 10 + \frac{25 - 13}{15} \times 10 = 18$ (1).	1+1+1
17	Highest frequency is 20, so the modal class is 40–50 (1); $l = 40$, $f_1 = 20$, $f_0 = 16$, $f_2 = 4$, $h = 10$ (1); Mode $= l + \frac{f_1 - f_0}{2f_1 - f_0 - f_2} \times h = 40 + \frac{4}{20} \times 10 = 42$ (1).	1+1+1
	OR: Take assumed mean $a = 85$; deviations $d_i = x_i - a$: -20, -10, 0, 10, 20 (1); $\sum f_i = 40$ and $\sum f_i d_i = -360$ (1); Mean $= a + \frac{\sum f_i d_i}{\sum f_i} = 85 + \frac{-360}{40} = 76$ (1).	1+1+1
18	Cumulative frequencies: 2, 6, 8, 23, 25; $N = 25$, so $\frac{N}{2} = 12.5$ (1); Median class 40–50: $l = 40$, $cf = 8$, $f = 15$, $h = 10$ (1); Median $= l + \frac{\frac{N}{2} - cf}{f} \times h = 40 + \frac{12.5 - 8}{15} \times 10 = 43$ (1).	1+1+1
	OR: Highest frequency is 16, so the modal class is 10–20 (1); $l = 10$, $f_1 = 16$, $f_0 = 8$, $f_2 = 8$, $h = 10$ (1); Mode $= l + \frac{f_1 - f_0}{2f_1 - f_0 - f_2} \times h = 10 + \frac{8}{16} \times 10 = 15$ (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	$\sum f_i = 21 + p$ and $\sum f_i x_i = 1535 + 75p$ (2); $\frac{1535 + 75p}{21 + p} = 73.4$ gives $(75 - 73.4)p = 6.4$ (2); $p = 4$ (1).	2+2+1
	OR: Median class is 30–40 with $f = 16$, and the cumulative frequency before it is $cf = 18 + x$ (2); $36.25 = 30 + \frac{50 - (18 + x)}{16} \times 10$ gives $x = 22$ (2); $x = 22$; then $y = 100 - 57 - x = 21$ (1).	2+2+1

20	$\sum f_i = 38 + p$ and $\sum f_i x_i = 2540 + 65p$ (2); $\frac{2540+65p}{38+p} = 66.4$ gives $(65 - 66.4)p = -16.8$ (2); $p = 12$ (1).	2+2+1
	OR: Median class is 70–80 with $f = 16$, and the cumulative frequency before it is $cf = 44 + x$ (2); $71.25 = 70 + \frac{50-(44+x)}{16} \times 10$ gives $x = 4$ (2); $x = 4$; then $y = 100 - 76 - x = 20$ (1).	2+2+1

Mistakes to avoid:

- Taking cf of the median class as its own cumulative frequency; cf in the formula is that of the class *before* the median class.
- Choosing the median class by comparing with n instead of $\frac{n}{2}$.
- Confusing f_0 and f_2 (preceding and succeeding class frequencies) in the mode formula.
- Using the class limits instead of the class marks x_i in $\sum f_i x_i$.
- In the step-deviation method, forgetting to multiply $\frac{\sum f_i u_i}{\sum f_i}$ by h.
- Using the modal class (highest frequency) as the median class, or vice versa.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapter 13: **Statistics** | Class 10 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** 8	$\frac{990+25p}{42+p} = 23.8$ gives $(25-23.8)p = 9.6$; $p = 8$
2	**(c)** $\frac{x_i - a}{h}$	Definition of u_i
3	**(d)** 21.6	$\sum f_i = 50$ and $\sum f_i x_i = 1080$; Mean $= \frac{\sum f_i x_i}{\sum f_i} = \frac{1080}{50} = 21.6$
4	**(b)** 34	Median class 30–40: $l = 30$, $cf = 21$, $f = 10$, $h = 10$; Median $= l + \frac{\frac{N}{2} - cf}{f} \times h = 30 + \frac{25-21}{10} \times 10 = 34$
5	**(b)** 33.75	$l = 30$, $f_1 = 17$, $f_0 = 8$, $f_2 = 2$, $h = 10$; Mode $= l + \frac{f_1 - f_0}{2f_1 - f_0 - f_2} \times h = 30 + \frac{9}{24} \times 10 = 33.75$
6	**(a)** 60–70	It belongs to the class 60–70; Modal class = 60–70
7	**(c)** 10	$\frac{730+35p}{40+p} = 21.6$ gives $(35-21.6)p = 134$; $p = 10$
8	**(c)** cumulative frequencies	Median class from cumulative frequency
9	**(a)** 82.4	$\sum f_i = 100$ and $\sum f_i x_i = 8240$; Mean $= \frac{\sum f_i x_i}{\sum f_i} = \frac{8240}{100} = 82.4$
10	**(a)** 41	Median class 40–50: $l = 40$, $cf = 12$, $f = 5$, $h = 10$; Median $= l + \frac{\frac{N}{2} - cf}{f} \times h = 40 + \frac{12.5-12}{5} \times 10 = 41$

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Highest frequency is 14, so the modal class is 70–80; $l = 70$, $f_1 = 14$, $f_0 = 10$, $f_2 = 2$, $h = 10$ (1); Mode $= l + \frac{f_1 - f_0}{2f_1 - f_0 - f_2} \times h = 70 + \frac{4}{16} \times 10 = 72.5$ (1).	1+1
12	Cumulative frequencies: 9, 30, 48, 68, 86, 100; $N = 100$, so $\frac{N}{2} = 50$; Median class 70–80: $l = 70$, $cf = 48$, $f = 20$, $h = 10$ (1); Median $= l + \frac{\frac{N}{2} - cf}{f} \times h = 70 + \frac{50-48}{20} \times 10 = 71$ (1).	1+1
13	Class marks x_i: 25, 35, 45, 55, 65; $\sum f_i = 40$ and $\sum f_i x_i = 1810$ (1); Mean $= \frac{\sum f_i x_i}{\sum f_i} = \frac{1810}{40} = 45.25$ (1).	1+1
14	Highest frequency is 19, so the modal class is 50–60; $l = 50$, $f_1 = 19$, $f_0 = 5$, $f_2 = 5$, $h = 10$ (1); Mode $= l + \frac{f_1 - f_0}{2f_1 - f_0 - f_2} \times h = 50 + \frac{14}{28} \times 10 = 55$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Take assumed mean $a = 45$; deviations $d_i = x_i - a$: -20, -10, 0, 10, 20 (1); $\sum f_i = 40$ and $\sum f_i d_i = 10$ (1); Mean $= a + \frac{\sum f_i d_i}{\sum f_i} = 45 + \frac{10}{40} = 45.25$ (1).	1+1+1
16	Cumulative frequencies: 2, 5, 12, 22, 25; $N = 25$, so $\frac{N}{2} = 12.5$ (1); Median class 40–50: $l = 40$, $cf = 12$, $f = 10$, $h = 10$ (1); Median $= l + \frac{\frac{N}{2} - cf}{f} \times h = 40 + \frac{12.5-12}{10} \times 10 = 40.5$ (1).	1+1+1
17	Highest frequency is 20, so the modal class is 20–30 (1); $l = 20$, $f_1 = 20$, $f_0 = 5$, $f_2 = 10$, $h = 10$ (1); Mode $= l + \frac{f_1 - f_0}{2f_1 - f_0 - f_2} \times h = 20 + \frac{15}{25} \times 10 = 26$ (1).	1+1+1
	OR: Take assumed mean $a = 85$; deviations $d_i = x_i - a$: -20, -10, 0, 10, 20 (1); $\sum f_i = 40$ and $\sum f_i d_i = 0$ (1); Mean $= a + \frac{\sum f_i d_i}{\sum f_i} = 85 + \frac{0}{40} = 85$ (1).	1+1+1
18	Cumulative frequencies: 2, 13, 23, 31, 40; $N = 40$, so $\frac{N}{2} = 20$ (1); Median class 80–90: $l = 80$, $cf = 13$, $f = 10$, $h = 10$ (1); Median $= l + \frac{\frac{N}{2} - cf}{f} \times h = 80 + \frac{20-13}{10} \times 10 = 87$ (1).	1+1+1
	OR: Highest frequency is 21, so the modal class is 90–100 (1); $l = 90$, $f_1 = 21$, $f_0 = 20$, $f_2 = 18$, $h = 10$ (1); Mode $= l + \frac{f_1 - f_0}{2f_1 - f_0 - f_2} \times h = 90 + \frac{1}{4} \times 10 = 92.5$ (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	$\sum f_i = 39 + p$ and $\sum f_i x_i = 1415 + 25p$ (2); $\frac{1415+25p}{39+p} = 33.8$ gives $(25-33.8)p = -96.8$ (2); $p = 11$ (1).	2+2+1

	OR: Median class is 80–90 with $f = 8$, and the cumulative frequency before it is $cf = 18 + x$ (2); $83.75 = 80 + \frac{25-(18+x)}{8} \times 10$ gives $x = 4$ (2); $x = 4$; then $y = 50 - 30 - x = 16$ (1).	2+2+1
20	$\sum f_i = 20 + p$ and $\sum f_i x_i = 850 + 45p$ (2); $\frac{850+45p}{20+p} = 43$ gives $(45 - 43)p = 10$ (2); $p = 5$ (1).	2+2+1
	OR: Median class is 60–70 with $f = 16$, and the cumulative frequency before it is $cf = 20 + x$ (2); $67.5 = 60 + \frac{50-(20+x)}{16} \times 10$ gives $x = 18$ (2); $x = 18$; then $y = 100 - 63 - x = 19$ (1).	2+2+1

Mistakes to avoid:

- Taking cf of the median class as its own cumulative frequency; cf in the formula is that of the class *before* the median class.
- Choosing the median class by comparing with n instead of $\frac{n}{2}$.
- Confusing f_0 and f_2 (preceding and succeeding class frequencies) in the mode formula.
- Using the class limits instead of the class marks x_i in $\sum f_i x_i$.
- In the step-deviation method, forgetting to multiply $\frac{\sum f_i u_i}{\sum f_i}$ by h.
- Using the modal class (highest frequency) as the median class, or vice versa.

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ANSWER KEY & MARKING SCHEME SET 3

Chapter 13: **Statistics** | Class 10 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** 65	$l=60$, $f_1=10$, $f_0=5$, $f_2=5$, $h=10$; Mode $=l+\frac{f_1-f_0}{2f_1-f_0-f_2}\times h=60+\frac{5}{10}\times 10=65$
2	**(a)** 80–90	It belongs to the class 80–90; Modal class = 80–90
3	**(b)** 11	$\frac{7415+75p}{89+p}=82.4$ gives $(75-82.4)p=-81.4$; $p=11$
4	**(c)** with the highest frequency	Definition
5	**(c)** 35	$\sum f_i=25$ and $\sum f_ix_i=875$; Mean $=\frac{\sum f_ix_i}{\sum f_i}=\frac{875}{25}=35$
6	**(c)** 37.5	Median class 30–40: $l=30$, $cf=8$, $f=16$, $h=10$; Median $=l+\frac{\frac{N}{2}-cf}{f}\times h=30+\frac{20-8}{16}\times 10=37.5$
7	**(c)** 63	$l=60$, $f_1=15$, $f_0=12$, $f_2=8$, $h=10$; Mode $=l+\frac{f_1-f_0}{2f_1-f_0-f_2}\times h=60+\frac{3}{10}\times 10=63$
8	**(a)** 60–70	It belongs to the class 60–70; Modal class = 60–70
9	**(c)** 2	$\frac{825+25p}{23+p}=35$ gives $(25-35)p=-20$; $p=2$
10	**(d)** $a+\frac{\sum f_id_i}{\sum f_i}$	Assumed mean method

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Cumulative frequencies: 4, 20, 24, 32, 47, 50; $N=50$, so $\frac{N}{2}=25$; Median class 100–110: $l=100$, $cf=24$, $f=8$, $h=10$ (1); Median $=l+\frac{\frac{N}{2}-cf}{f}\times h=100+\frac{25-24}{8}\times 10=101.25$ (1).	1+1
12	Class marks x_i: 65, 75, 85, 95, 105; $\sum f_i=40$ and $\sum f_ix_i=3400$ (1); Mean $=\frac{\sum f_ix_i}{\sum f_i}=\frac{3400}{40}=85$ (1).	1+1
13	Highest frequency is 14, so the modal class is 60–70; $l=60$, $f_1=14$, $f_0=12$, $f_2=8$, $h=10$ (1); Mode $=l+\frac{f_1-f_0}{2f_1-f_0-f_2}\times h=60+\frac{2}{8}\times 10=62.5$ (1).	1+1
14	Cumulative frequencies: 5, 13, 28, 42, 50; $N=50$, so $\frac{N}{2}=25$; Median class 90–100: $l=90$, $cf=13$, $f=15$, $h=10$ (1); Median $=l+\frac{\frac{N}{2}-cf}{f}\times h=90+\frac{25-13}{15}\times 10=98$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Take assumed mean $a=95$; deviations $d_i=x_i-a$: -20, -10, 0, 10, 20 (1); $\sum f_i=50$ and $\sum f_id_i=-210$ (1); Mean $=a+\frac{\sum f_id_i}{\sum f_i}=95+\frac{-210}{50}=90.8$ (1).	1+1+1
16	Cumulative frequencies: 3, 10, 30, 37, 50; $N=50$, so $\frac{N}{2}=25$ (1); Median class 90–100: $l=90$, $cf=10$, $f=20$, $h=10$ (1); Median $=l+\frac{\frac{N}{2}-cf}{f}\times h=90+\frac{25-10}{20}\times 10=97.5$ (1).	1+1+1
17	Highest frequency is 8, so the modal class is 40–50 (1); $l=40$, $f_1=8$, $f_0=4$, $f_2=4$, $h=10$ (1); Mode $=l+\frac{f_1-f_0}{2f_1-f_0-f_2}\times h=40+\frac{4}{8}\times 10=45$ (1).	1+1+1
	OR: Take assumed mean $a=85$; deviations $d_i=x_i-a$: -30, -20, -10, 0, 10, 20 (1); $\sum f_i=50$ and $\sum f_id_i=-390$ (1); Mean $=a+\frac{\sum f_id_i}{\sum f_i}=85+\frac{-390}{50}=77.2$ (1).	1+1+1
18	Cumulative frequencies: 4, 19, 23, 27, 40; $N=40$, so $\frac{N}{2}=20$ (1); Median class 20–30: $l=20$, $cf=19$, $f=4$, $h=10$ (1); Median $=l+\frac{\frac{N}{2}-cf}{f}\times h=20+\frac{20-19}{4}\times 10=22.5$ (1).	1+1+1
	OR: Highest frequency is 8, so the modal class is 80–90 (1); $l=80$, $f_1=8$, $f_0=4$, $f_2=4$, $h=10$ (1); Mode $=l+\frac{f_1-f_0}{2f_1-f_0-f_2}\times h=80+\frac{4}{8}\times 10=85$ (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	$\sum f_i=32+p$ and $\sum f_ix_i=1200+55p$ (2); $\frac{1200+55p}{32+p}=41$ gives $(55-41)p=112$ (2); $p=8$ (1).	2+2+1

	OR: Median class is 70–80 with $f = 20$, and the cumulative frequency before it is $cf = 30 + x$ (2); $78 = 70 + \frac{50-(30+x)}{20} \times 10$ gives $x = 4$ (2); $x = 4$; then $y = 100 - 74 - x = 22$ (1).	2+2+1
20	$\sum f_i = 36 + p$ and $\sum f_i x_i = 2250 + 85p$ (2); $\frac{2250+85p}{36+p} = 64.75$ gives $(85 - 64.75)p = 81$ (2); $p = 4$ (1).	2+2+1
	OR: Median class is 30–40 with $f = 5$, and the cumulative frequency before it is $cf = 13 + x$ (2); $32 = 30 + \frac{25-(13+x)}{5} \times 10$ gives $x = 11$ (2); $x = 11$; then $y = 50 - 23 - x = 16$ (1).	2+2+1

Mistakes to avoid:

- Taking cf of the median class as its own cumulative frequency; cf in the formula is that of the class *before* the median class.
- Choosing the median class by comparing with n instead of $\frac{n}{2}$.
- Confusing f_0 and f_2 (preceding and succeeding class frequencies) in the mode formula.
- Using the class limits instead of the class marks x_i in $\sum f_i x_i$.
- In the step-deviation method, forgetting to multiply $\frac{\sum f_i u_i}{\sum f_i}$ by h.
- Using the modal class (highest frequency) as the median class, or vice versa.

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